Problem of Interest

Which cities among the potential candidate cities selected by the company should be allocated robotaxis, and how many?

To organize my thoughts, I tried to set a formula starting from simple models to developing into complex models. **The main model I suggest is the last model.**

① Basic newsvendor model,
② Model with added costs
③ Model that maximizes profit
④ multi-penalties.

All of the formulations below determine city selection and vehicle numbers simultaneously.

All demand, throughput, and costs use the same evaluation period. After the evaluation period is set, and the model determines the city/vehicle deployment to apply during that period. (Need to think about how to set the period.)

1. Notation used in common across every model

Symbol	Meaning	How is the value determined?
C	Set of potential candidate cities	The company's candidate list.
S	Set of demand scenarios	Need to be generated
y_c	1 if city c is entered, else 0	**Determined by the optimization**
q_c	Number of vehicles deployed in city c	**Determined by the optimization**
D_c^s	Potential robotaxi demand in scenario s	**Estimated from data + assumptions for missing parts**
κ_c	Number of trips one vehicle can serve during the period	Operational data or assumptions or fixed.
Q	Total number of vehicles available for deployment	Company input. If unavailable, set an experimental value
$\bar{q}_c$	Upper bound on vehicle deployment in each city	Facility and operating limits. Q if there is no separate limit
a_c	1 if operation in the city is feasible at the planning stage	Assessed using permit, ODD, and facility data. Separate assumptions for unverified conditions
B	Upper bound on the number of cities that can be entered simultaneously	Company input or an experimental value **only when an actual restriction exists**
R_c^s	Actual throughput	A value calculated based on the deployment outcome in the model
O_c^s	Remaining service capacity	A value calculated based on the deployment outcome in the model
U_c^s	Unserved potential demand	A value calculated based on the deployment outcome in the model

D_c^s is a sampled value of uncertain demand. **After the scenarios are generated, it is a fixed input during optimization**, and the model does not choose demand.

2. Common constraints

$$
\begin{aligned}
y_c \le q_c \le \bar{q}_c y_c \qquad & \forall c \in C, \\
\sum_{c \in C} q_c \le Q, \quad \sum_{c \in C} y_c \le B & \\
y_c \le a_c \qquad & \forall c \in C, \\
y_c \in \{0,1\}, \quad q_c \in \mathbb{Z}_{\ge 0}. &
\end{aligned}
$$

Constraint	Meaning
$y_c \le q_c \le \bar{q}_c y_c$	Zero vehicles if the city is not entered; at least one vehicle is deployed if it is entered
$\sum_c q_c \le Q$	Total vehicle limit
$\sum_{c\in C} y_c \le B$	Limit on the number of cities selected
$y_c \le a_c$	Enter only cities where operation is feasible

3. Values calculated based on the deployment outcome

Once demand D_c^s and service capacity $\kappa_c q_c$ are determined, the following three values are calculated based on the deployment outcome; **they are not variables that need to be investigated separately.**

$$
\begin{aligned}
R_c^s &= \min\{D_c^s, \kappa_c q_c\} && \text{Actual throughput,}\\
O_c^s &= (\kappa_c q_c - D_c^s)^+ && \text{Remaining service capacity,}\\
U_c^s &= (D_c^s - \kappa_c q_c)^+ && \text{Unserved potential demand.}
\end{aligned}
$$

Here, $(x)^+ = \max\{x, 0\}$. All three values are measured in trips.

① Basic form: Minimize the problems of surplus vehicles and shortages

$$
\min_{y,q} \frac{1}{|S|} \sum_{s\in S} \sum_{c\in C} \left[\underbrace{O_c^s}_{\text{Remaining capacity}} + \underbrace{\rho U_c^s}_{\text{Loss from shortages}} \right]
$$

Additional symbol	Meaning	How to set it
$\rho > 0$	How much more important one unit of shortage is considered than one unit of remaining capacity	**Researcher assumption or company preference.** E.g., sensitivity experiments comparing 1, 2, and 5

This model has no need of precise company cost data, but excludes entry costs of cities.

② Cost form: Minimize total costs, including entry and vehicle costs

$$
\begin{aligned}
\min_{y,q} \quad & \underbrace{\sum_c F_c y_c}_{\text{City entry costs}} + \underbrace{\sum_c K_c q_c}_{\text{Vehicle deployment and holding costs}} \\
& + \frac{1}{|S|} \sum_s \sum_c \left[\underbrace{c_c^+ O_c^s}_{\text{Loss from remaining capacity}} + \underbrace{c_c^- U_c^s}_{\text{Opportunity loss from unserved demand}} \right]
\end{aligned}
$$

Additional symbol	Meaning and unit	How to set it / Required data
F_c	Fixed cost of entering the city, in KRW or dollars	**Additional data required:** Costs of establishing a base and initial preparation. Explicit assumptions if unavailable
K_c	Deployment and holding cost per vehicle over the evaluation period	**Additional data required:** Vehicle, insurance, fixed operating costs, etc.
c_c^+	Additional loss per trip of remaining service capacity	Company data or assumptions. **Do not double-count the same costs as K_c**
c_c^-	Loss per trip of missed potential demand	Data or assumptions on contribution margin, customer churn, etc.

③ Profit form: Maximize the profit from served rides

$$\max_{y,q} \quad \frac{1}{|S|}\sum_s \sum_c \underbrace{m_c R_c^s}_{\text{Contribution margin from served rides}} - \underbrace{\sum_c F_c y_c}_{\text{Entry costs}} - \underbrace{\sum_c K_c q_c}_{\text{Fixed vehicle costs}}$$

Additional symbol	Meaning	How to set it / Required data
m_c	Revenue from one trip minus the variable costs incurred for that trip	**Additional data required:** Fares and variable costs such as operation and energy. Assumptions if unavailable

If customers cannot be served, the profit that would be earned decreases. **Adding the same revenue loss again as $c_c^- U_c^s$ may result in double counting.**

This most directly reflects the company's perspective, but fare and variable-cost data are required. If it is not profitable, the result may be to enter no city.